

HAT-P-37b

R-2533

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_180613 · created 2026-08-25T23:21:53+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458282.922 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.045
Transit depth (Rp/Rs)^2	2.46 +/- 1.4 [%]
Orbital Inclination (inc)	89.88 +/- 2.19
Airmass coefficient 1 (a1)	1.319 +/- 0.054
Airmass coefficient 2 (a2)	-0.26 +/- 0.22
Scatter in the residuals of the lightcurve fit is	4.06 %
Best Comparison Star	#3 - [277, 329]
Optimal Aperture	3.53
Optimal Annulus	7.74
Transit Duration (day)	0.0879 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.157 $\pm$ 0.045 vs 0.1394 (deviation 0.25 σ)
Duration fit vs published	0.0879 d vs 0.0983 d (deviation 0.98 σ)
Mid-time O-C	-10.1 min
Depth SNR	2.2
Residual scatter	4.06 %

Light curve

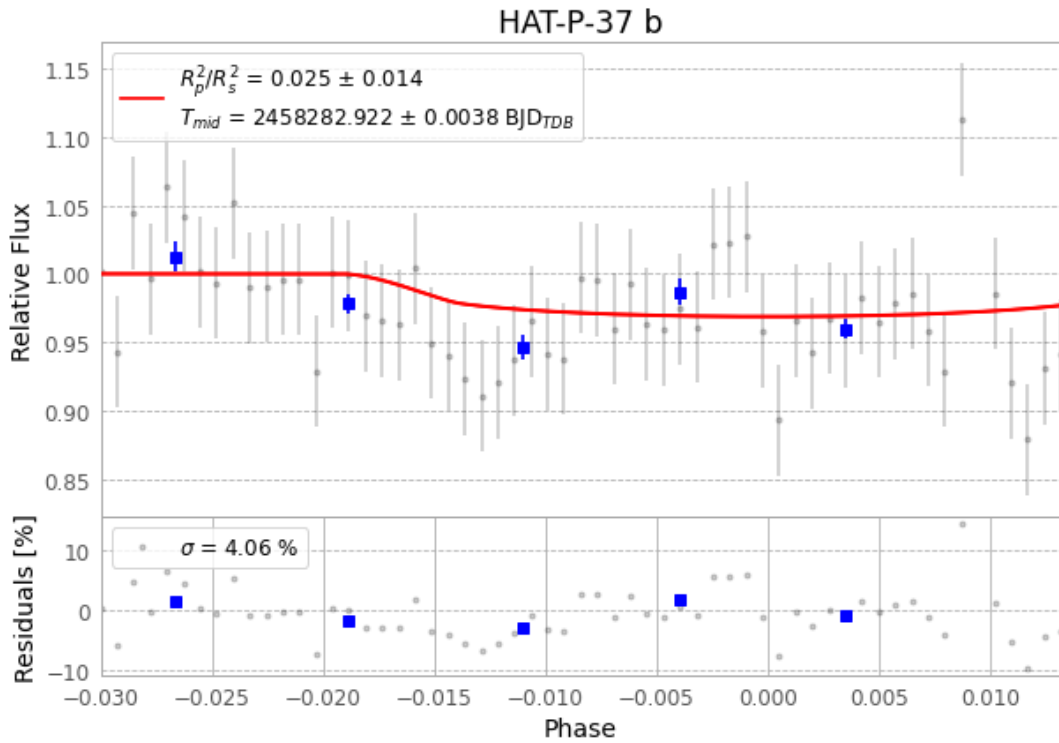


Figure 1 — FinalLightCurve_HAT-P-37 b_2018-06-13.png (SHA-256 39a5532beda61144...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 269], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 89e910211a6a4b98...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), HATP-37180613080312.FITS → HATP-37180613111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-180613.log (d4ab8650590c...), FinalLightCurve_HAT-P-37 b_2018-06-13.png (39a5532beda6...), FinalLightCurve_HAT-P-37 b_2018-06-13.csv (c114e9ec3707...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:21Z.